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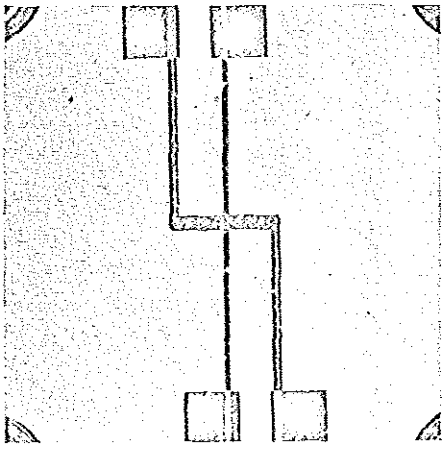
THE JOSEPHSON EFFECTS

Predicted in 1962 by a British graduate student, these two unusual manifestations of the superconducting state have now been observed directly. Their theoretical and technological significance is discussed

by Donald N. Langenberg, Douglas J. Scalapino and Barry N. Taylor

Four years ago Brian D. Josephson, a young graduate student in physics at the University of Cambridge, made a startling prediction. On the basis of a purely theoretical analysis of the phenomenon of superconductivity (the abrupt disappearance of electrical resistance in certain substances at temperatures near absolute zero) Josephson came to the conclusion that in principle a "supercurrent" consisting of correlated pairs of electrons could be made to flow across an insulating gap between two superconducting bodies, provided that the gap was small enough. He further suggested that this "tunneling" of electron pairs through an insulator could take two forms, which have come to be known as the Josephson effects. Both forms have been observed in recent experiments.

The nature of the Josephson effects can be understood in a general way by considering what happens when an electric current flows in a superconductor [*see illustration on page 32*]. If a direct current is made to flow in a bar consisting of a superconducting substance, a voltmeter connected across the ends of the bar will indicate that the drop in voltage across the bar is zero; thus the bar has no resistance and is said to be in the superconducting state. If the bar is now divided in two and the pieces are separated by a distance of, say, a centimeter, no current will flow in the opened circuit, and the voltmeter will indicate a voltage equal to the open-circuit voltage of the current source (for example a battery).



JOSEPHSON JUNCTION consists of two superconducting tin strips separated by an insulating layer of tin oxide 10 angstroms thick. The glass slide on which the junction is mounted is about an inch square. The procedure for fabricating such junctions is shown in the bottom illustration on page 34.

If the distance between the two pieces is reduced to about 10 angstrom units, however, one of two surprising things can happen. The first is that the voltmeter shows no voltage between the two pieces of superconductor, but a direct current is observed to flow in the circuit even though the two pieces are still physically separated. In other words, the current can flow without resistance not only through the two pieces of superconductor but also across the gap between them. This phenomenon is called the d.c. Josephson effect.

The second thing that can happen is that a direct current flows in the circuit, but the voltmeter indicates a voltage. Simultaneously very high-frequency electromagnetic radiation emanates from the gap, indicating the presence in the gap of a very high-frequency alternating current. This phenomenon is called the a.c. Josephson effect. Like the d.c. effect, the a.c. effect is a direct consequence of the unique nature of the superconducting state.

The investigation of the two effects has already made important contributions to our understanding of superconductivity, and it now appears that they may also have useful applications in technology. Such applications include devices for generating extremely short-wavelength electromagnetic radiation, for measuring very small magnetic fields and for providing highly precise voltage references. Perhaps more important, the Josephson effects may enable physicists to gain an even deeper understanding of superconductivity and to measure some of the fundamental constants of physics with an accuracy never before achieved. Before discussing in detail these somewhat exotic phenomena and their connection with superconductivity, we should like to review some of the salient features of the superconducting state.

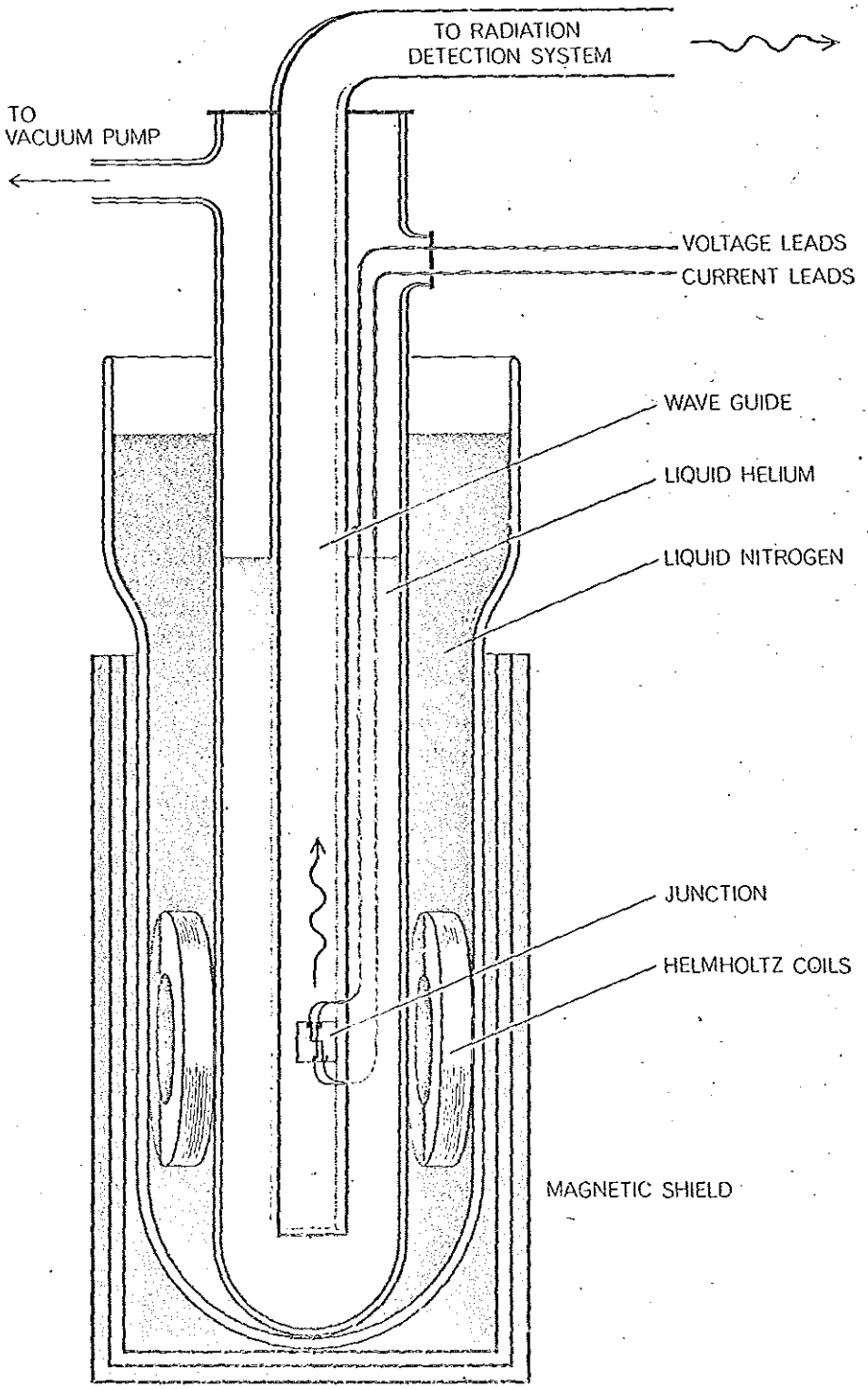
The phenomenon of superconductivity was discovered in 1911 by Heike Kamerlingh Onnes of the Netherlands. He observed that in a sample of mercury all resistance to the flow of electric current vanished when the sample was cooled below a characteristic temperature (now called the superconducting transition temperature). Since 1911 the number of known superconductors has grown to several hundred and includes a number of metallic elements, many metallic alloys and even a few semiconductors. The transition temperatures of all the known superconductors lie below 20 degrees Kelvin (degrees centigrade above absolute zero), and in most cases the transition temperatures are below 5 degrees K., a temperature region that is accessible only by using liquid helium as a refrigerant or by resorting to more sophisticated techniques. The experimental study of superconductivity therefore lies in the realm of low-temperature physics.

For several decades after Onnes' discovery it was believed that a metal in

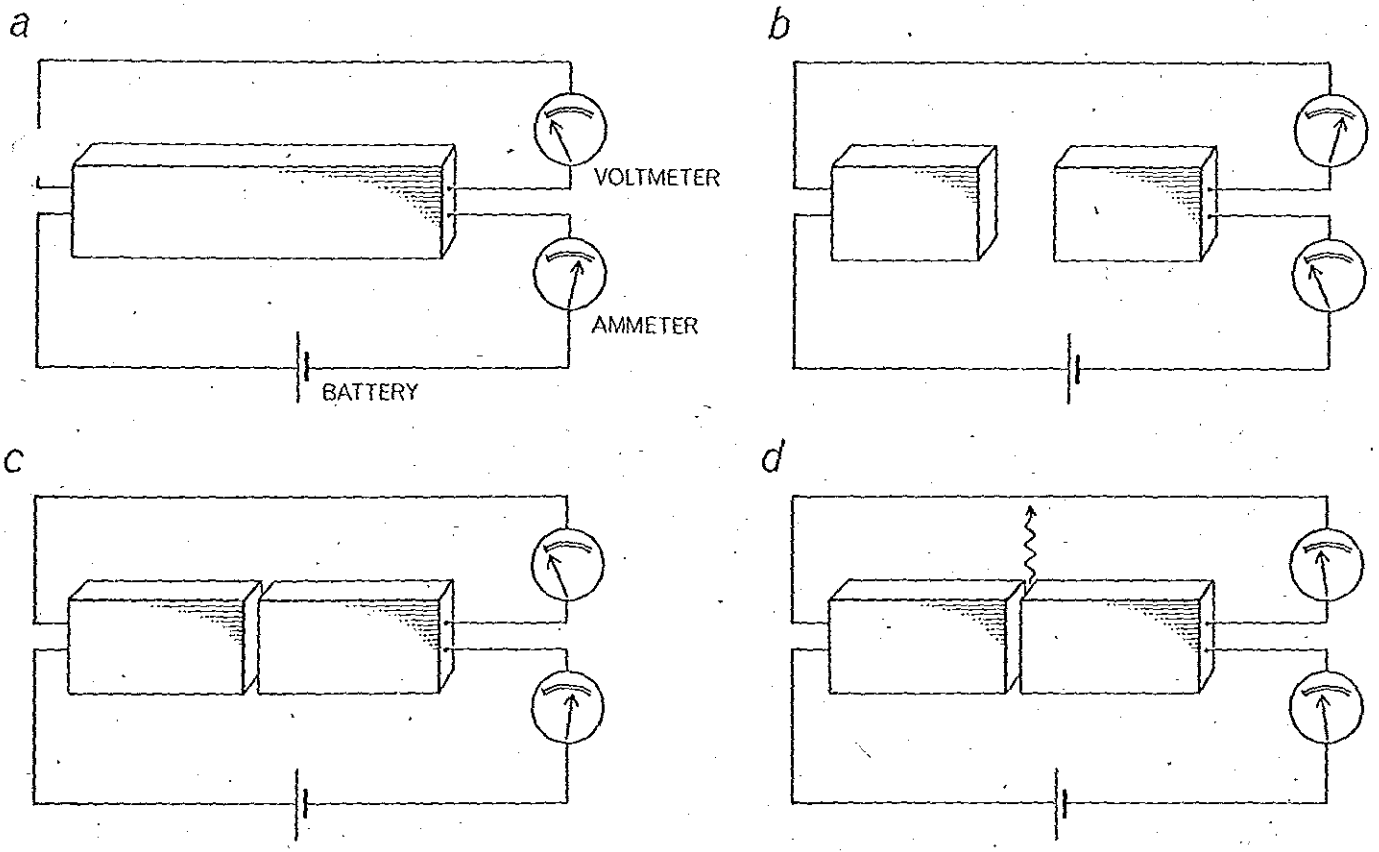
the superconducting state could be characterized simply as a metal with zero electrical resistance or infinite electrical conductivity. Then in 1933 the German investigators W. Meissner and R. Ochsenfeld discovered that there is more to it than that. They found that an externally applied magnetic field is excluded from a superconductor either when it is cooled below its transition temperature in the presence of the field or when the field is applied after the substance has become superconducting. A substance with infinite conductivity would exhibit the latter behavior but not the former; it would tend to "freeze in" any field present when it became superconducting, not expel it. This effect, now called the Meissner effect, is therefore not a consequence of infinite conductivity but a distinct and perhaps more fundamental property of the superconducting state.

The existence of the Meissner effect and certain other properties of superconductors led the theoretical physicist Fritz London to a highly significant insight into the nature of superconductivity. He suggested in 1935 that the superconducting state is a manifestation of quantum mechanics operating on a macroscopic scale—a state with long-range order, in which the motions of a significant fraction of the electrons throughout the entire bulk of the superconductor are correlated. Thus a piece of metal in the superconducting state is in some respects like a single giant molecule. This idea led to a qualitative understanding of many of the unique properties of superconductors. It suggested, for example, an explanation for the characteristic infinite conductivity. A normal metal exhibits resistance to the flow of electric current because any concerted motion of the electrons in one direction is dissipated rapidly by the scattering of individual electrons from the vibrating atoms of the metal or from impurities or imperfections in its structure. If the electrons are in a state in which the motion of each electron is correlated with the motions of all the other electrons, however, and if this correlation extends throughout the metal, the scattering of one electron will necessarily involve all the others. This "united we stand, divided we fall" property makes the scattering of electrons impossible or at least highly improbable, and a supercurrent flow, once started, is immune to the scattering that retards normal currents.

Although London's idea provided physicists with a new and fruitful microscopic origin and nature of the superconducting state was not really understood until 1957, when John Bardeen, L. N. Cooper and J. R. Schrieffer, then at the University of Illinois, published a now famous theoretical paper on superconductivity. According to their theory, superconductivity occurs because of the existence of an attractive force between the electrons in a metal. Although electrons, each having a negative charge, repel each other in free space, this need not be the case in a metal. There a negatively charged elec-



EXPERIMENTAL SETUP was used by the authors to detect microwave radiation emitted by Josephson junctions. The junctions were mounted inside a wave guide: a rectangular metal tube in which electromagnetic radiation can propagate. Wires were connected to the junction so that a current could be supplied to the junction and the voltage across the junction could be measured. The wave guide and the junction were inserted in a Dewar vessel (a vacuum-insulated container similar to a Thermos bottle) that could be filled with liquid helium for cooling the junction to a temperature one or two degrees above absolute zero. Outside the helium Dewar vessel there was a pair of coils for producing the necessary magnetic fields. The entire arrangement was surrounded by a triple-walled magnetic shield to eliminate the earth's magnetic field inside the Dewar vessel.



JOSEPHSON EFFECTS can be understood in a general way by considering what happens when an electric current flows in a superconductor. The absence of resistance in a superconductor is demonstrated by the fact that no voltage drop appears across a superconducting bar when a direct current is made to flow through it (*a*). If the bar is divided in two and the pieces are separated by a distance of, say, a centimeter (*b*), no current will flow in the opened circuit, and the voltmeter will indicate a voltage equal to the open-circuit voltage of the current source (in this case a battery). If the distance between the two pieces is reduced to about 10 angstroms, however, one of two surprising things can happen. The first is that the voltmeter shows no voltage between the two pieces of superconductor but a direct current is observed to flow in the circuit even though the two pieces are still physically separated (*c*). This phenomenon is called the d.c. Josephson effect. The second thing that can happen is that a direct current flows in the circuit but the voltmeter indicates a voltage (*d*). Simultaneously very high-frequency radiation emanates from the gap, indicating the presence in the gap of a very high-frequency alternating current. This phenomenon is called the a.c. Josephson effect.

tron moving through a lattice of metal ions attracts the positively charged ions. This distorts the lattice, creating a wake of excess positive charge to which another electron can be attracted. Thus in a metal there can be, in addition to the usual repulsive force between electrons, an indirect attractive force that arises from the lattice of metal ions. In order for a metal to be a superconductor this lattice-mediated attractive force must exceed the repulsive force—the net interaction of electrons must be attractive.

The involvement of the lattice in the attractive interaction explains the seemingly peculiar fact that superconductivity has never been observed in the metals that are usually considered to be the best conductors, such as copper and silver, whereas it is a common phenomenon among the poorer conductors, such as lead and tin. The high conductivity of copper and silver is a consequence of the comparatively weak interaction of their electrons with the lattice in these single electrons that impairs conductivity in the normal, or nonsuperconducting, state, but it also reduces the attractive interaction of electrons that leads to superconductivity.

The attractive force tends to couple together in "bound pairs" electrons that have equal and opposite momentum and spin. This binding is extremely weak, however, as revealed by the fact that it is disrupted, and superconductivity is destroyed, by thermal disorder at temperatures just a few degrees above absolute zero. Because the attraction is weak the two paired electrons are, on the average, separated by a distance that is thousands of times greater than the distance between the lattice ions. Since there are usually several electrons for each ion, the electrons of each bound pair range over a volume that simultaneously contains millions of other electron pairs. This spatial overlapping of the pairs has important consequences that arise primarily from the principle of quantum mechanics called the exclusion principle, which states that two electrons with the same spin cannot occupy the same spatial position. If the requirements of the exclusion principle are to be met, the motions of all the pairs must be correlated. The Bardeen-Cooper-Schrieffer theory showed that the correlation is accomplished if the centers of mass of all the pairs move with the same momentum. This equality of the center-of-mass momentum of the bound electron pairs throughout the superconductor is the long-range order envisioned by London. All the electron pairs are locked together by the requirement that they have a common center-of-mass momentum. Disturbing one pair disturbs all the pairs, because of the cooperative nature of this momentum-ordering.

In the quantum-mechanical description of this ordering the wave aspects of electrons must be taken into account. An electron pair with a center-of-mass momentum p can be described by a wave with wavelength h/p, where h is Planck's constant. In terms of the quantum-mechanical wave picture, equality

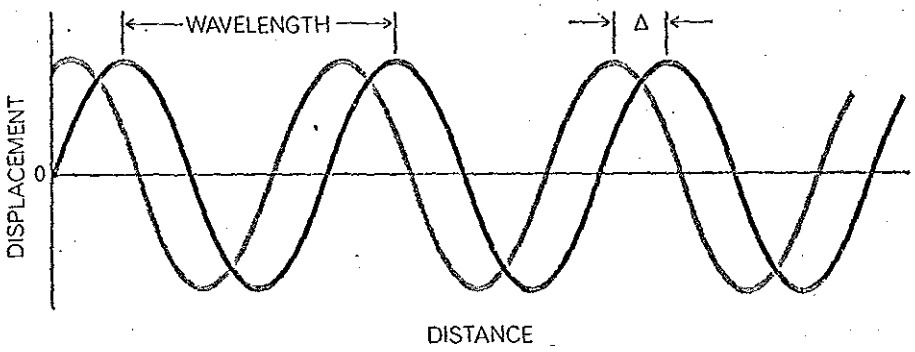
of the center-of-mass momentum of all the pairs means that the waves describing the pairs have the same wavelength. There is more to the superconducting correlations than this, however. The upper illustration on this page shows two waves that have the same wavelength but are not identical. They can be made identical by sliding the light-colored wave forward by a certain distance. The waves are then said to be in phase, or to have the same phase. In a superconductor the waves of all the pairs have not only the same wavelength but also the same phase.

To reiterate, the concept of the phase of an electron pair is a purely quantum-mechanical notion arising from the wave properties of matter. The identity of the pair phases in a superconductor is a quantum-mechanical effect operating on a macroscopic scale. This phase identity is enforced by energy considerations. In order to optimize the average binding energy of the spatially overlapping pairs within the restrictions imposed by the exclusion principle, the phases of all pair waves must be the same. This is the precise quantum-mechanical statement corresponding to the fact that the center-of-mass momentum of all pairs is the same. Because this is the key to understanding the Josephson effects, we must be quite explicit about what we mean by phase.

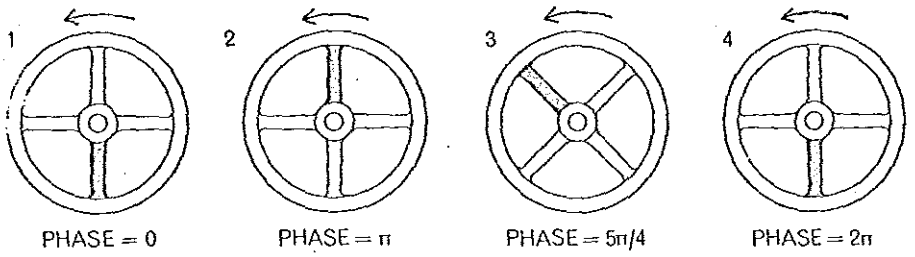
The phase of a periodic motion is a measure of the fractional number of cycles (multiplied by 2π) through which the system has advanced from some arbitrary but fixed reference configuration. An example that may clarify the meaning of phase is a rotating wheel with one colored spoke [*see lower illustration on this page*]. Taking as a reference the position in which the colored spoke is vertically down (*position 1*), the phase is indicated for several different configurations. For example, in position 2 the wheel has turned through half a cycle, and we say the phase has advanced by π. It is clear, comparing position *1* and position *4*, that the physical configuration of the wheel is the same for a phase equal to zero and a phase equal to 2π. We say that the phase of position *1* and position *4* are the same, with respect to the "modulus" 2π. The motion of the wheel depends on the way the phase develops in time. For example, if the phase of the wheel varies linearly with time, the wheel rotates at constant speed. The time interval during which the wheel completes a cycle and the phase increases by 2π is called the period of the motion.

In general, a system can be periodic in space as well as in time. An example is the displacement of the surface of a pond when a wave is created by dropping in a pebble. The phase of such a wave depends on both the spatial distance and the time lapse with respect to some reference point in space and time. In the top illustration on the next page the space and time development of a simple sinusoidal wave is shown. The wavelength of the disturbance is the distance over which one spatial oscillation occurs and the phase of the displacement changes by 2π. The period is the time interval over which one temporal oscillation occurs and the phase of the displacement changes by 2π. The phase at several points on the wave is given with respect to the point of origin. The broken line shows the space-time development of a particular point of constant phase, the one corresponding to zero phase. The rate at which this point of constant phase moves in space is called the phase velocity of the wave.

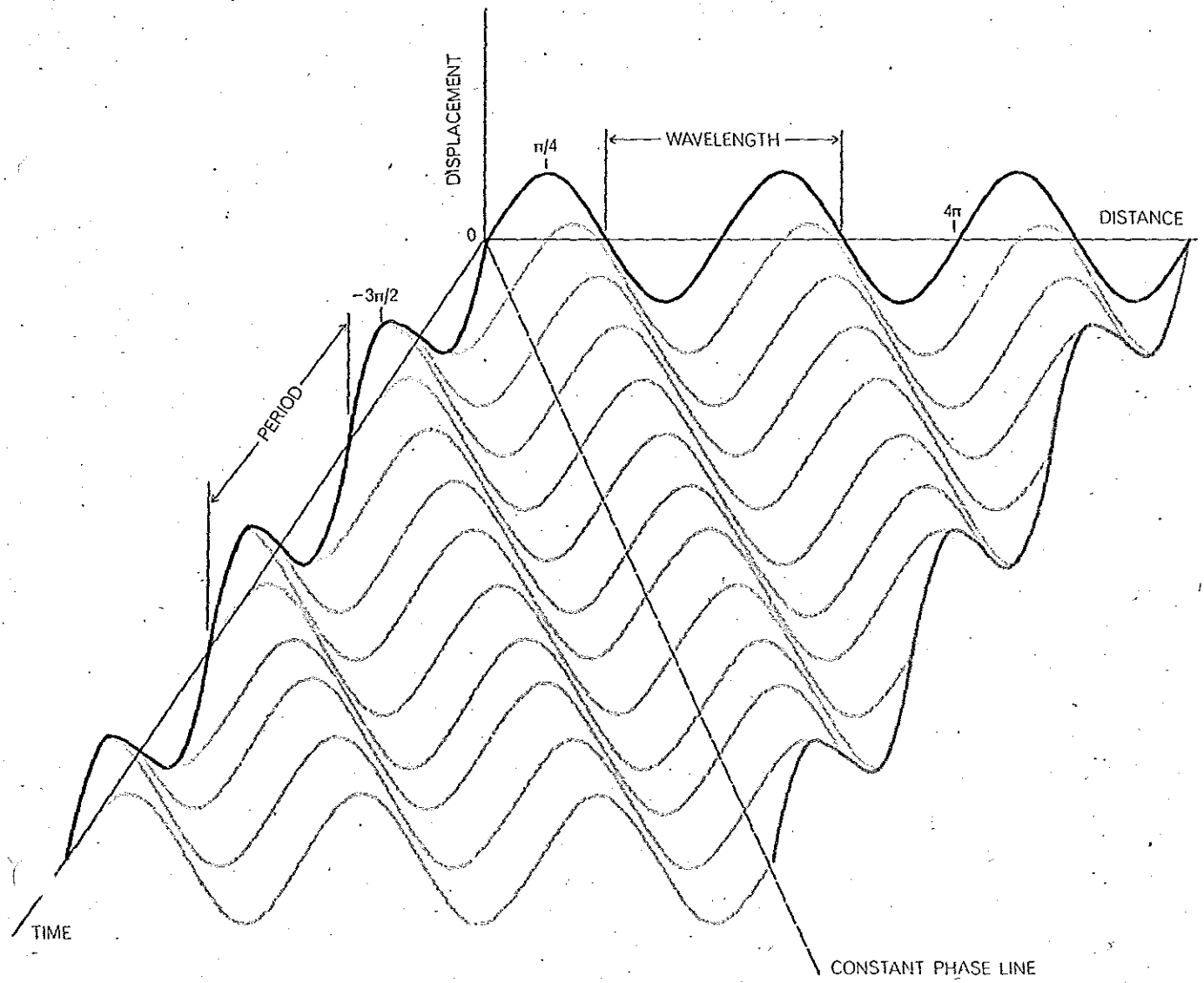
Now, as we have mentioned, according to quantum mechanics the wave associated with an electron pair has a spatial oscillation whose wavelength is determined by the center-of-mass momentum of the pair. The wave also has a time oscillation whose period is determined by the energy of the electron pair, so that the phase of the pair depends on the pair's energy as well as on its center-of-mass momentum. As we have seen, in a bulk superconductor the attractive interaction of electrons, the large number of bound electron pairs and the exclusion principle conspire to make the pair phases the same. These two facts lead to a simple explanation of the zero resistance of the superconducting state: If a voltage difference V existed between the two ends of the current-carrying superconducting bar shown in the illustration on the opposite page, the energy of an electron pair in one end would be greater by $2eV$ than the energy of an electron pair in the other end (e is the charge of an electron). Then, as time passed, a phase difference would develop between the electron pairs in the two ends of the bar. Rather than go to the higher energy state that this breakdown of the pair phase-locking would produce, the superconductor carries the current with-



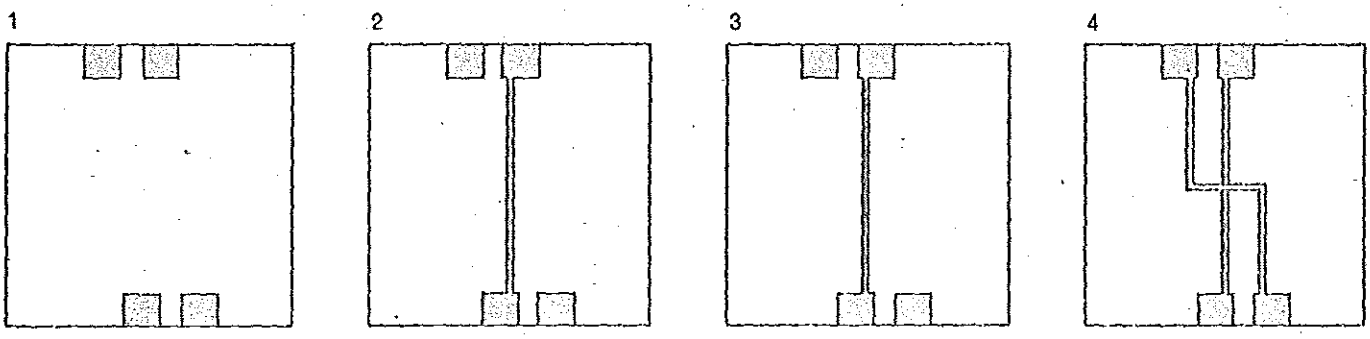
TWO WAVES can have the same wavelength but different phases. They can be made identical by sliding the light-colored wave forward by the distance Δ. The waves are then said to be in phase, or to have the same phase. In a superconductor the waves of all the electron pairs have not only the same wavelength but also the same phase.



ROTATING WHEEL can be used to clarify the meaning of phase. The phase of the wheel's motion is a measure of the fractional number of cycles (multiplied by 2π) through which the wheel has advanced from some arbitrary but fixed reference configuration. Taking as a reference the position in which the colored spoke is vertically down (*position 1*), the phase is given for three other configurations. The phase of position *1* and position *4* are said to be the same with respect to the "modulus" 2π. The time interval during which the wheel completes a cycle and the phase is increased by 2π is called the period of the motion.



TRAVELING WAVE, that is, one that varies periodically in both space and time, is shown here for a simple sinusoidal disturbance analogous to the ripples produced on the surface of a pond by dropping in a pebble. The wavelength is the distance over which one spatial oscillation occurs. The period is the time interval over which one temporal oscillation occurs. The phase at several points on the wave is given with respect to point of origin (*0*). Broken line shows space-time development of a particular point of constant phase, the one corresponding to zero phase. The rate at which this point moves in space is called the phase velocity of the wave. According to quantum mechanics the wave associated with an electron pair in a superconductor has a spatial oscillation, whose wavelength is determined by the pair's center-of-mass momentum, and a time oscillation, whose period is determined by the pair's energy.



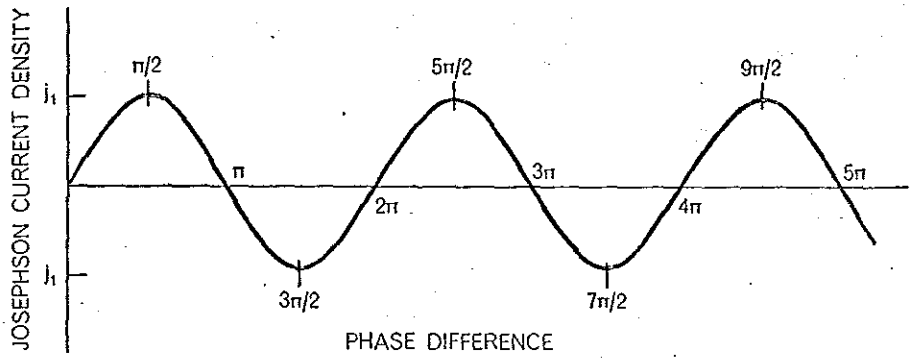
PROCEDURE for fabricating a Josephson junction begins with the deposition of four electrodes, in the form of thin metal films, on a slide (*1*). This is done by heating the metal in a vacuum chamber until some of it evaporates; the resulting metallic vapor is passed through a mask before being deposited on the slide. Next a thin film of a superconducting substance (for example tin) is deposited on the slide in the form of a strip that makes contact with two of the previously deposited electrodes (*2*). The strip is typically one millimeter wide and about 2,000 angstroms thick. The third step is the formation of a very thin layer (about 10 angstroms thick) of insulating oxide on the surface of the bottom strip (*3*). This is accomplished by admitting a carefully controlled amount of air or pure oxygen to the vacuum chamber. The junction is completed by depositing a second strip of superconductor on top of the oxide layer, usually at right angles to the bottom strip (*4*). This strip makes contact with the two remaining electrodes. Wires for supplying a current to the junction and for measuring the voltage drop across the junction are then soldered to the electrodes.

out allowing a voltage difference to appear.

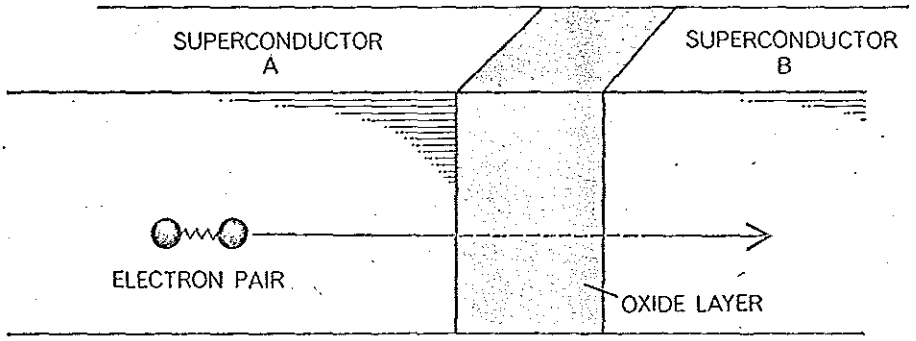
What Josephson did in 1962 was to perceive that it should be possible to modify the phase-locking and hence study in detail this aspect of the superconducting long-range order. He consjdered the situation shown in the illustration on page 32 and argued as follows: The pair phase must be the same throughout a single piece of superconductor, since pairs can move freely through it. If the superconductor is divided into two pieces and the pieces are well separated so that pairs cannot be exchanged between them, there need be no particular relation between the pair phases of the two pieces. If the two superconductors can be brought close enough together, however, an intermediate situation will result in which there can be some exchange of electron pairs by means of the quantum-mechanical process called tunneling. (Because of their wavelike nature, electrons can tunnel, or penetrate, through barriers they could not penetrate if they were simply particles.) In this case the overall system can assume a state of minimum energy in which there is a unique difference between the pair phases in the two superconductors. It should then be possible to vary this phase difference, or relative phase, by controlling the rate of pair transfer between the two pieces of superconductor.

The intermediate situation can in principle be achieved simply by separating the two pieces by a distance on the order of 10 angstroms. Unfortunately no one actually knows how to construct such an arrangement, but a completely equivalent one can be realized in practice using a procedure developed by Ivar Giaever and John C. Fisher of the General Electric Research Laboratory [*see bottom illustration on opposite page*]. The essential feature of the technique is that the two superconductors are separated by a thin layer of insulating oxide, which is allowed to form on one of them before the other is deposited on top. The result is a metal-insulator-metal sandwich with superconducting bread and insulating salami. Such a structure is called a tunnel junction [*see illustration on page 30*]. The tunneling of pairs between the two superconductors of such a junction leads to a coupling of the pair phases on the two sides, which can be modified by electric and magnetic fields as Josephson envisioned.

Josephson predicted that if somehow a phase difference is produced between

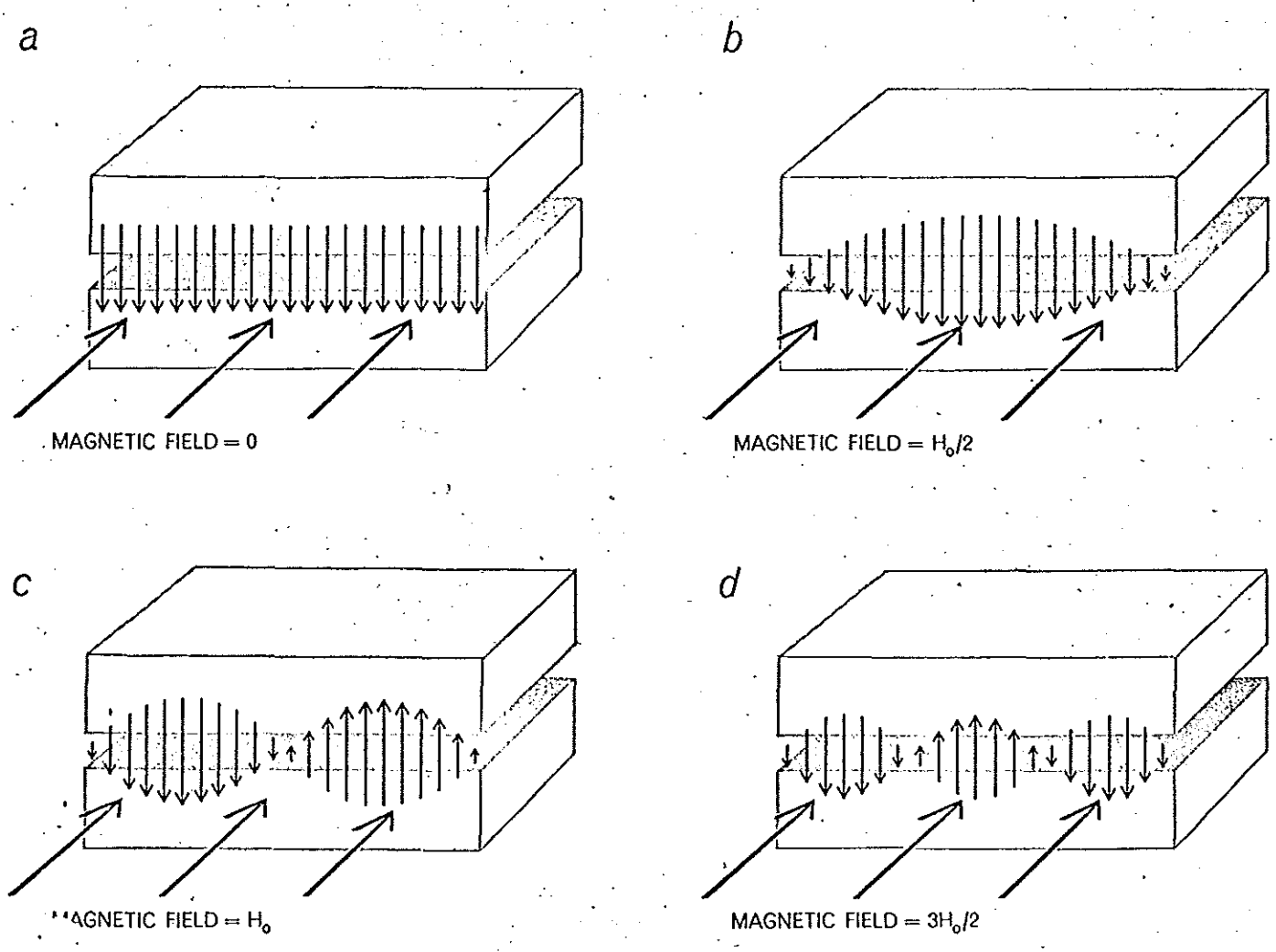


CURRENT DENSITY, or current per unit area, in a Josephson junction is a sinusoidal function of the difference in the phases of the electron pairs in the two superconductors. Current density has a maximum value of j_1 when the phase difference is some odd multiple of $\pi/2$. Current density is zero when the phase difference is an integer multiple of π.



PHASE OF PAIRS IN A	PHASE OF PAIRS IN B	CHANGE IN PHASE THROUGH OXIDE LAYER	PHASE OF PAIR		JOSEPHSON CURRENT (MAXIMUM EQUALS j_1)
			AFTER GOING FROM A TO B	AFTER GOING FROM B TO A	
0	0	0	0	0	0
0	π/4	π/4	π/4	π/2	$j_1/\sqrt{2}$
0	π/2	π/2	π/2	π	j_1
0	3π/4	3π/4	3π/4	3π/2	$j_1\sqrt{2}$
0	π	π	π	2π	0

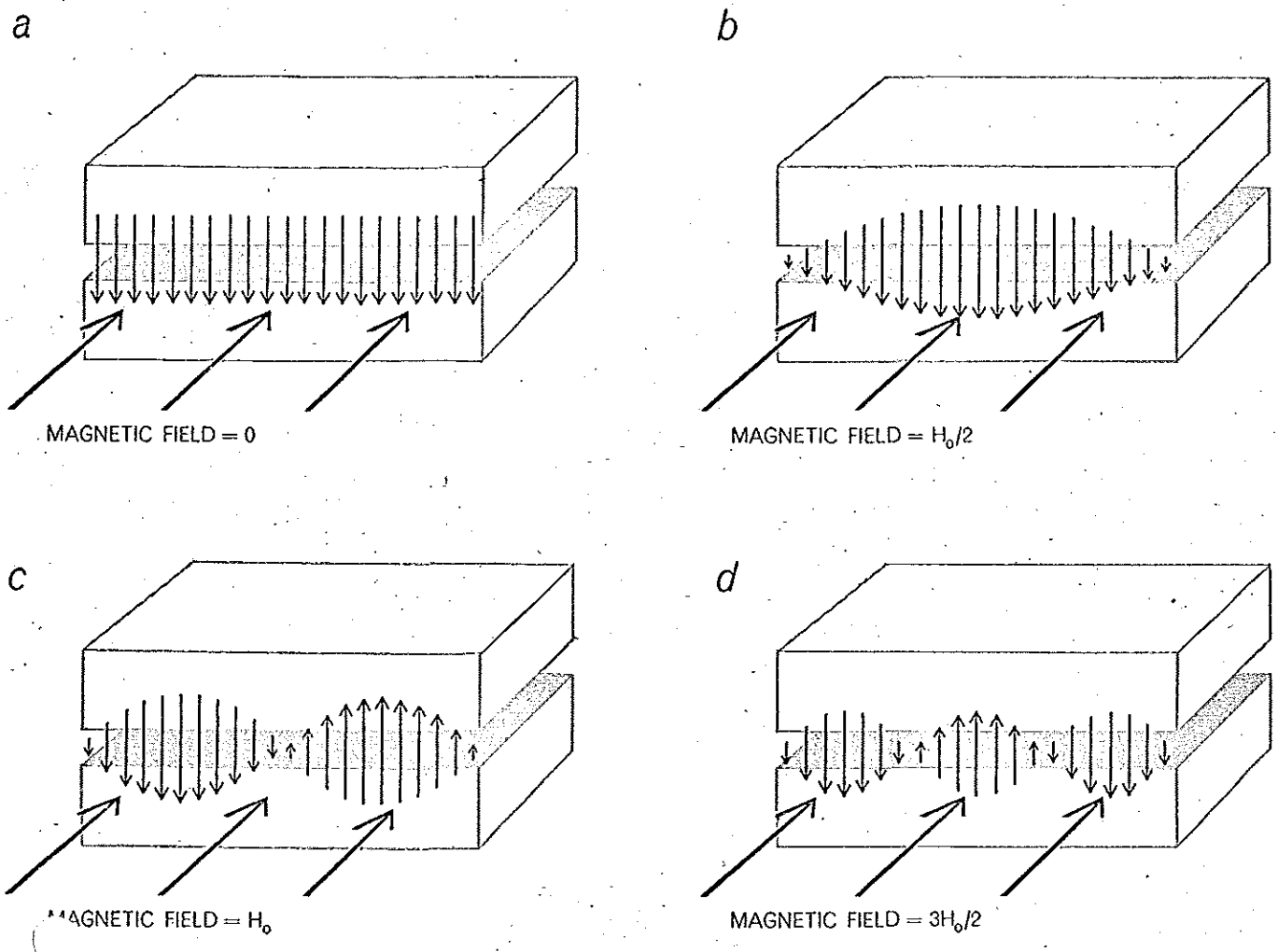
PERIODIC RELATION between the junction current density and the pair phase difference can be understood by remembering that the phase of a pair depends on its motion. The schematic diagram at top shows two superconductors, labeled *A* and *B*, separated by an insulating oxide layer. The table at bottom shows how the transfer of electron pairs from *A* to *B* and from *B* to *A* is related to the difference in the pair phases in the two superconductors; it also indicates the net current for this relative pair phase. The pair phase in superconductor *A* is taken to be zero. If the pair phase in superconductor *B* is also zero (*first line in table*), the transfer of pairs in either direction is equally likely and the net junction current will be zero. In the second line of the table the situation for a pair phase of $\pi/4$ in superconductor *B* is shown. The change in phase of a pair that traverses the oxide layer is $\pi/4$. The phase of a pair going from *A* to *B* thus changes from zero to $\pi/4$; when it reaches side *B*, its phase exactly matches the phase on side *B*, it fits in nicely with the other pairs on side *B*, and such a transfer is relatively probable. The phase of a pair going from *B* to *A*, however, changes from $\pi/4$ to $\pi/2$, which does not match well with the zero pair phase of the other pairs on side *A*. The transfer of a pair from *B* to *A* is therefore not as likely to occur as the transfer of a pair from *A* to *B*. On the average there will be a net flow of pairs from *A* to *B* and hence a current. The phase mismatch for pairs going from *B* to *A* in the third line of the table is even worse, and therefore the number of pairs going from *B* to *A* is less. Consequently the net current from *A* to *B* is increased. As the relative pair phase continues to increase in lines four and five, the mismatch is reduced and net current from *A* to *B* decreases again. Table describes first half-cycle of curve at top of page.



EFFECT OF A MAGNETIC FIELD on a direct current passing through a Josephson junction is shown in these schematic diagrams. The external magnetic field (*black arrows*) is applied in the plane of the junction at right angles to one side of the junction. In *a* the magnetic field is zero and the magnitude of the current, as represented by the length of the colored arrows, is the same at all points of the junction. In *b* the applied magnetic field ($H_0/2$) is just strong enough to cause the wavelength of the Josephson current to equal twice the length of the junction; thus half the wavelength fits exactly into the length of the junction. In *c* the magnetic field has been increased to H_0, so that a full wavelength fits into the junction. The current now reverses direction once; equal currents flow in both directions and the total junction current is zero. In *d* the field has been increased to $3H_0/2$; three half-wavelengths fit into the junction, the current reverses direction twice and again there is a net current across the junction. The total current is not as large as it was in *a* or *b*. A graph showing how the total junction current depends on the strength of the magnetic field appears on the opposite page.

the two superconductors, pairs will be preferentially transferred in one direction and a current will flow between the two superconductors. (In a real laboratory experiment it is the current that is under the direct control of the experimenter, not the phase difference. The phase difference is produced by causing a current to flow in the junction. It is, however, conceptually simpler to think of the phase difference as the controllable quantity and then to ask what current results.) The junction current density in Josephson's theory is a sinusoidal function of the phase difference [*see top illustration on preceding page*]. This rather strange periodic relation between the current density and the phase can be understood by remembering that the phase of a pair depends on its motion. Pairs transferred in the preferred direction are simply changing their phase by just the amount required to match the pair phase on the side to which they are going. Pairs that attempt to travel in the opposite direction will shift their phase by the same amount. If the resulting phase matches the pair phase on the side to which they are going, they are able to make the transition, the number of pairs traveling in both directions is the same and the net current is zero. If the resulting phase does not match, the pairs will be reflected back to the side from which they came, and the transfer of pairs in the preferred direction will give rise to a current [*see bottom illustration on preceding page*]. There is thus a direct current flow without a voltage drop between the two superconductors; the *entire* junction behaves like a single bar of superconductor. This is the d.c. Josephson effect.

The first indisputable experimental confirmation of the d.c. Josephson effect was reported in 1963 by John Rowell of the Bell Telephone Laboratories. He observed the predicted resistanceless current flow in a junction and studied its dependence on an externally applied magnetic field. Observation of the magnetic-field dependence was essential, because such a resistanceless current could also be due to the presence of small "bridges" of superconductor across the junction. (Indeed, it appears in retrospect that many workers had observed the d.c. Josephson effect but had ascribed the resistanceless currents they observed to such bridges and had done their best to ignore this supposed undesirable "dirt" effect.) A bridge current can be modified only by a rather large magnetic field, but the d.c. Josephson current has



EFFECT OF A MAGNETIC FIELD on a direct current passing through a Josephson junction is shown in these schematic diagrams. The external magnetic field (*black arrows*) is applied in the plane of the junction at right angles to one side of the junction. In *a* the magnetic field is zero and the magnitude of the current, as represented by the length of the colored arrows, is the same at all points of the junction. In *b* the applied magnetic field ($H_0/2$) is just strong enough to cause the wavelength of the Josephson current to equal twice the length of the junction; thus half the wavelength fits exactly into the length of the junction. In *c* the magnetic field has been increased to H_0, so that a full wavelength fits into the junction. The current now reverses direction once; equal currents flow in both directions and the total junction current is zero. In *d* the field has been increased to $3H_0/2$; three half-wavelengths fit into the junction, the current reverses direction twice and again there is a net current across the junction. The total current is not as large as it was in *a* or *b*. A graph showing how the total junction current depends on the strength of the magnetic field appears on the opposite page.

the two superconductors, pairs will be preferentially transferred in one direction and a current will flow between the two superconductors. (In a real laboratory experiment it is the current that is under the direct control of the experimenter, not the phase difference. The phase difference is produced by causing a current to flow in the junction. It is, however, conceptually simpler to think of the phase difference as the controllable quantity and then to ask what current results.) The junction current density in Josephson's theory is a sinusoidal function of the phase difference [*see top illustration on preceding page*]. This rather strange periodic relation between the current density and the phase can be understood by remembering that the phase of a pair depends on its motion. Pairs transferred in the preferred direction are simply changing their phase by just the amount required to match the pair phase on the side to which they are going. Pairs that attempt to travel in the opposite direction will shift their phase by the same amount. If the resulting phase matches the pair phase on the side to which they are going, they are able to make the transition, the number of pairs traveling in both directions is the same and the net current is zero. If the resulting phase does not match, the pairs will be reflected back to the side from which they came, and the transfer of pairs in the preferred direction will give rise to a current [*see bottom illustration on preceding page*]. There is thus a direct current flow without a voltage drop between the two superconductors; the *entire* junction behaves like a single bar of superconductor. This is the d.c. Josephson effect.

The first indisputable experimental confirmation of the d.c. Josephson effect was reported in 1963 by John Rowell of the Bell Telephone Laboratories. He observed the predicted resistanceless current flow in a junction and studied its dependence on an externally applied magnetic field. Observation of the magnetic-field dependence was essential, because such a resistanceless current could also be due to the presence of small "bridges" of superconductor across the junction. (Indeed, it appears in retrospect that many workers had observed the d.c. Josephson effect but had ascribed the resistanceless currents they observed to such bridges and had done their best to ignore this supposed undesirable "dirt" effect.) A bridge current can be modified only by a rather large magnetic field, but the d.c. Josephson current has

a unique periodic variation in quite small magnetic fields. This variation arises in the following way.

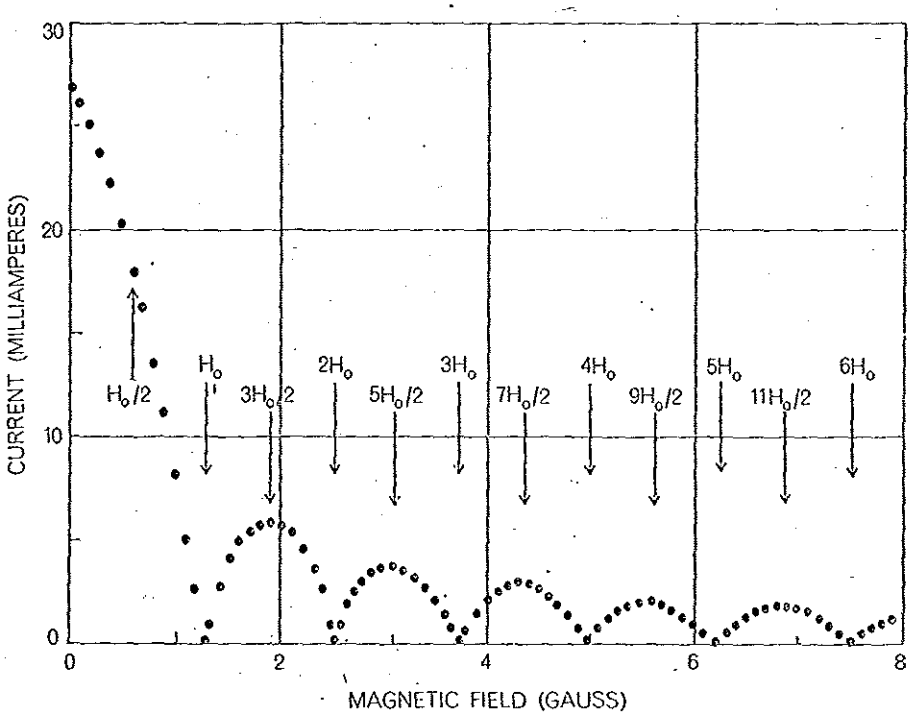
In quantum mechanics the phase of the wave associated with a particle (or a bound electron pair in a superconductor) depends on the magnetic field (more precisely, on something called the magnetic vector potential). It turns out that if there is a magnetic field in a tunnel junction, the difference between the pair phases in the two superconductors has a spatial variation. If the field happens to be uniform and constant throughout the junction oxide layer, the phase varies in the plane of the junction in a direction at right angles to the direction of the field and at a rate proportional to the magnitude of the field. Because of the periodic dependence of the current density on the phase, the direct current in the junction oscillates in space and may reverse its direction at several points in the junction if the magnetic field is large enough. The maximum net resistanceless direct current that can be carried by the junction therefore varies periodically as the magnetic field is increased [*see illustration on this page*]. Rowell's observation of precisely this behavior clinched the identification of the observed resistanceless current as the d.c. Josephson effect.

So far we have considered only situations where the pair phase difference between the two superconductors of the junction does not vary in time. We have seen, however, that if there is a voltage difference and hence a potential-energy difference between the two superconductors, the phase difference *will* vary in time. The time-independent phase difference that exists when a direct current flows in the junction is in fact initially produced by a momentary voltage difference across the junction that occurs when an external current source is connected to it. The voltage drop causes the phase to shift in time. When the phase difference attains the value that corresponds to the current determined by the current source, the voltage drop vanishes, and the relative pair phase thereafter remains constant in time. This happens in about a ten-billionth of a second. If the current source forces the current to exceed the maximum d.c. Josephson current, however, a voltage appears across the junction, and the relative phase increases steadily in time. Because of the sinusoidal dependence of the Josephson current on the phase, the current oscillates back and forth between the two superconductors at a frequency proportional to the voltage across the junction. This is the a.c. Josephson effect.

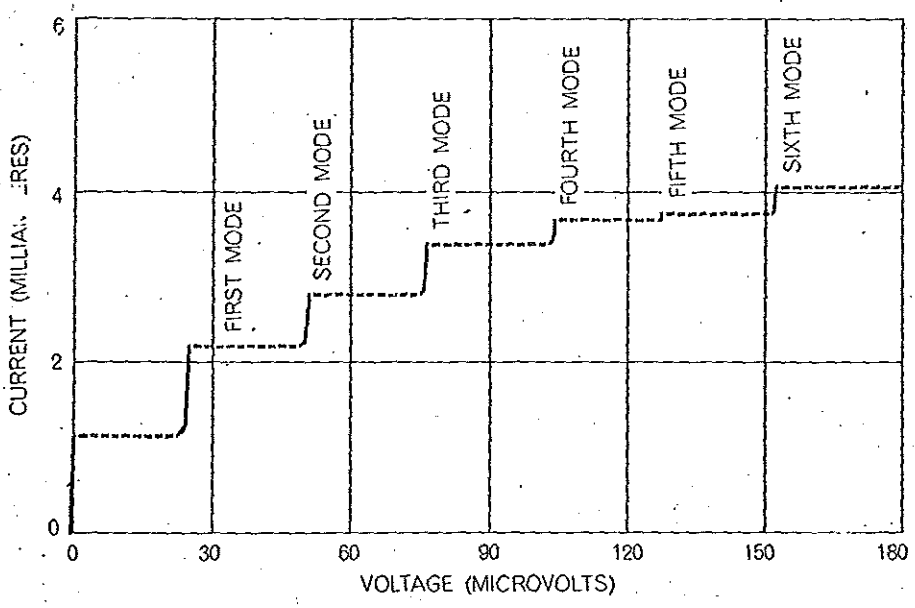
Experimental confirmation of the a.c. Josephson effect has proved to be considerably more difficult than confirmation of the d.c. effect. The relation between the time dependence of the phase and the voltage across the junction is such that if the voltage has a constant value V, the frequency ν of the oscillating supercurrent is $2e/h$ times the voltage, where e is the charge of an electron and h is Planck's constant (thus $\nu = 2eV/h$). The ratio $2e/h$ is numerically equal to 483.6 megacycles per microvolt, and since typical junction voltages range from a few microvolts to several millivolts the frequency of the oscillating supercurrent can be as high as several hundred billion cycles per second. The obvious straightforward way to observe such high-frequency currents is to detect the electromagnetic waves radiated by the current. (In general any oscillating current will radiate electromagnetic waves.) At the frequencies involved the radiation is in the microwave and far-infrared regions of the electromagnetic spectrum. The amount of energy the a.c. Josephson currents radiate, however, is very small and thus rather difficult to detect. Consequently direct experimental confirmation of the a.c. Josephson effect came only after several indirect experiments had left little doubt that the effect really existed.

The first of these indirect confirmations was achieved in 1963 by S. Shapiro, then working for Arthur D. Little, Inc. Shapiro varied the voltage across the junction and measured the current through the junction while it was exposed to microwave radiation supplied by a conventional microwave oscillator. He found abrupt increases in the junction current at certain voltages. When these voltages were substituted into the Josephson frequency-voltage equation, $\nu = 2eV/h$, the frequencies turned out to be whole-number multiples of the frequency of the applied microwave radiation.

This effect can be explained as follows: In addition to a d.c. voltage across the junction, there is also a small voltage induced by the applied microwave radiation that oscillates at the frequency of this radiation. Since the frequency of the Josephson supercurrent depends on the junction voltage, the supercurrent frequency has a small periodic variation in time; in other words, the supercurrent is frequency-modulated. Because of this it contains components that have many frequencies. These frequencies are the algebraic sums of the frequency of the applied microwave radiation plus (or minus) the frequency



MAXIMUM NET DIRECT CURRENT that can be carried by a Josephson junction varies periodically as the magnetic field is increased. The values of the magnetic fields shown in the diagrams on the opposite page appear in the graph. The observation of precisely this behavior in 1963 by John Rowell of the Bell Telephone Laboratories clinched the identification of the observed resistanceless current in the junction as the d.c. Josephson effect.



"STEPS," or abrupt increases, in junction current were observed in the curves that related the current to the voltage of Josephson junctions, even in the absence of externally applied microwave radiation. This observation was an indirect confirmation of the existence of the a.c. Josephson effect. Each step corresponds to a "resonant mode" of the oscillating electromagnetic field that is present between the two superconducting strips in the junction.

of the Josephson supercurrent plus (or minus) all the harmonic frequencies associated with these two fundamental frequencies. At d.c. voltages for which the Josephson frequency is equal to the microwave frequency or some whole-number multiple of it, among all the possible net frequencies there is one that is equal to zero. This means that there is a zero frequency or direct current at those values of d.c. junction voltage for which this special relation between Josephson frequency and microwave frequency occurs. All of this depends, of course, on the existence of the oscillating Josephson supercurrent in the first place, so that Shapiro's observation represents an indirect confirmation of its existence.

Another indirect confirmation was provided by two of us (Scalapino and Taylor) and R. E. Eck at the University of Pennsylvania. It was observed that "steps," or abrupt increases, in junction current similar to those found by Shapiro appeared in the curves that related the current to the voltage of Josephson junctions, even in the absence of any externally applied microwave radiation [*see illustration above*]. To understand the origin of these steps it is useful to recall that associated with any structure of finite size in which waves can propagate there is a discrete set of "resonant modes," each one corresponding to a particular relation between the wavelength or frequency of the wave and the dimensions of the structure. For example, if a hollow tube in which sound waves can propagate is closed at one end and left open at the other, one has an organ pipe with a set of resonant modes corresponding to a fundamental resonant frequency and a whole series of overtones, or harmonic resonant frequencies. If the pipe is excited by some source of sound at a frequency equal to the fundamental or one of the overtones, an intense standing sound wave is set up in the pipe: the pipe resonates. If it is excited at some frequency other than the fundamental frequency or one of the overtone frequencies, the response is not nearly so great.

Now, a Josephson junction is essentially an electromagnetic organ pipe. Electromagnetic waves can propagate between the two superconductors, and consequently a junction will have a set of resonant electromagnetic modes. A relatively intense electromagnetic field can be excited in the junction if the frequency and wavelength of the exciting source are adjusted so that they match the frequency and wavelength of one of these resonant modes. The origin of the steps observed by Eck, Scalapino and Taylor is now clear. The Josephson supercurrent can be made to vary periodically in space with a wavelength determined by an applied magnetic field, and periodically in time with a frequency determined by the voltage across the junction. When the frequency and wavelength of the supercurrent are thus matched to the frequency and wavelength of one of the junction modes, a large electromagnetic field will be generated in the junction. This field can then react on the Josephson current in the same manner as Shapiro's applied microwave radiation to produce the observed steps in the current-voltage curve.

This discovery was important for two reasons. It provided additional indirect evidence for the existence of the a.c. Josephson effect, and it also provided a key to understanding how the a.c. supercurrent could be made to generate enough electromagnetic radiation in the junction to make possible direct external detection of the radiation. Armed with this knowledge, we embarked on an effort to achieve direct detection of radiation emitted by Josephson junctions. We have recently succeeded in doing so. The results of these experiments confirm in a very direct fashion the existence of the a.c. Josephson effect in superconductors and, perhaps more important, provide us with a powerful new tool for studying the effect.

In our experiments we chose to work at a frequency of about 10 billion cycles per second, because it is a commonly used microwave frequency and equipment and techniques for it are readily available and well known. The voltage corresponding to this frequency is about 20 microvolts. We therefore prepared junctions with the proper lengths so that the first or second resonant mode (or step in the current-voltage curve) occurred at 20 microvolts. These junctions were mounted inside a wave guide—a rectangular metal tube in which electromagnetic radiation can propagate. Wires were connected to the junction so that a d.c. voltage could be applied and the d.c. junction current could be measured. The wave guide and junction were inserted in a Dewar vessel (a vacuum-insulated container similar to a Thermos bottle) that could be filled with liquid helium for cooling the junction to a temperature one or two degrees above absolute zero. Outside the Dewar vessel there was a pair of coils for producing the necessary magnetic fields. The entire arrangement was surrounded by a magnetic shield to eliminate the earth's magnetic field inside the Dewar vessel [*see illustration on page 31*].

Radiation emitted by the junction traveled inside the wave guide up out of the Dewar vessel and into an electronic system that was designed to detect very small amounts of power. In one version this system could detect

as little as [illegible] watt, about the amount of power in visible light received by a human eye from a 100-watt light bulb 300 miles away. Such high sensitivity was necessary because, although we knew how to make the Josephson supercurrent generate radiation inside the junction, we had not completely solved the problem of getting the radiation out of the junction and into the wave guide. Radiation propagating in the junction is almost entirely reflected back into the junction when it reaches the ends, and very little (only about one part in 100,000) is radiated out into the wave guide.

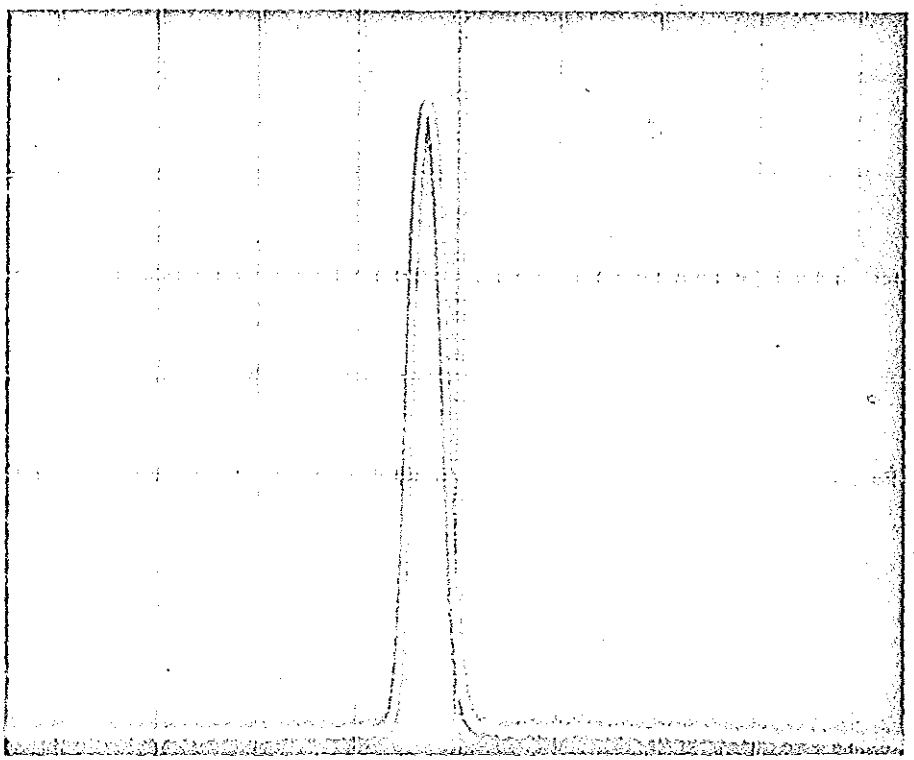
This problem of optimizing the transfer of power from a source (the junction) to a load (the wave guide and the detection system) is an impedance-matching problem analogous to the problem of efficiently transferring power from a high-impedance high-fidelity amplifier to a low-impedance loudspeaker or from a low-impedance man to a high-impedance boulder. The solutions in these examples are impedance-matching devices: an electrical transformer in the first case and a lever or a block and tackle in the second. The solution for the Josephson junction is not so obvious, and we elected to sidestep this difficulty for the moment simply by using a detection system with a sensitivity adequate for detecting the power that *was* radiated. We estimated we might get perhaps a trillionth (10^{-12}) of a watt; the available sensitivity of 10^{-10} watt provided some insurance.

Once radiation was detected we were able to improve the power transfer somewhat and succeeded in observing up to 10^{-11} watt. A typical signal curve appears in the illustration at the right. As predicted by Josephson, the radiation was found to be coherent (all in phase) and at least as monochromatic as the radiation emitted by a conventional vacuum-tube klystron oscillator. The ratio of the frequency of the radiation to the d.c. junction voltage was found to be equal to the predicted value of $2e/h$ to an accuracy of better than 1 percent. Experiments that turn out exactly as expected are rare, however, and this one was no exception. Along with these comforting agreements between theory and experiment there appeared some surprises and some new puzzles, which we are currently studying. Other workers are also active in the field. A group at the Ukrainian Academy of Sciences was the first to report a direct observation of radiation from a Josephson junction; they reported an observation of 10^{-13} watt of radiated power, and they are also carrying on a further study of the Josephson radiation.

The direct detection of the a.c. Josephson radiation has opened up the possibility of several rather diverse applications. It provides a sensitive and direct tool for further study of the Josephson effects and superconductivity. For example, simultaneous measurements of the d.c. junction voltage and the frequency of the emitted radiation can yield a highly accurate check on the validity of the Josephson frequency-voltage relation. This is of considerable importance because the frequency-voltage relation is directly connected with some of the most fundamental features of the present theory of superconductivity. If the relation holds up under such careful scrutiny, the fundamental physical constant ratio e/h that appears in it can be determined with an accuracy several times greater than has heretofore been achieved. Here is a unique opportunity to use a macroscopic quantum phenomenon to determine one of the fundamental constants of nature. Experiments along these lines are now in progress in our laboratory.

There are also possible applications in technology. The region of the electromagnetic spectrum between microwaves, with wavelengths measured in centimeters, and infrared waves, with wavelengths [illegible] of centimeters, is still relatively unexplored. It has been hard for physicists to perform experiments using radiation in this part of the spectrum because the generation and detection of such radiation has been difficult and expensive. The Josephson junction has promise as a simple and inexpensive source of small amounts of coherent and monochromatic radiation with wavelengths ranging from several millimeters down to a fraction of a millimeter. It would be particularly useful in low-temperature experiments where liquid-helium cooling is already an essential part of the experiment. It will be necessary to solve the power transfer problem in order to obtain the desired powers, but there appear to be no obstacles in principle to doing this.

It has been only four years since Josephson discerned in the theory of superconductivity the effects that now bear his name. During those years the effects have received abundant experimental confirmation and have provided striking direct evidence of the wondrous workings of quantum mechanics on a macroscopic scale in superconductors. Our newfound ability to control the phase of superconducting pairs promises new advances in our understanding of superconductivity as well as benefits in other areas of physics and technology.



OUTPUT SIGNAL produced by the detection system associated with the authors' experiment appears on the face of an oscilloscope tube. The signal was produced by an input signal of a hundred-billionth of a watt of microwave radiation from a Josephson junction.

JOSEPHSON EFFECT

III LAB

COPY 1.

DO NOT REMOVE FROM LA

About eight years ago a 22-year-old student showed his professor some calculations on quantum tunneling. Now that professor tells us

How Josephson discovered his effect

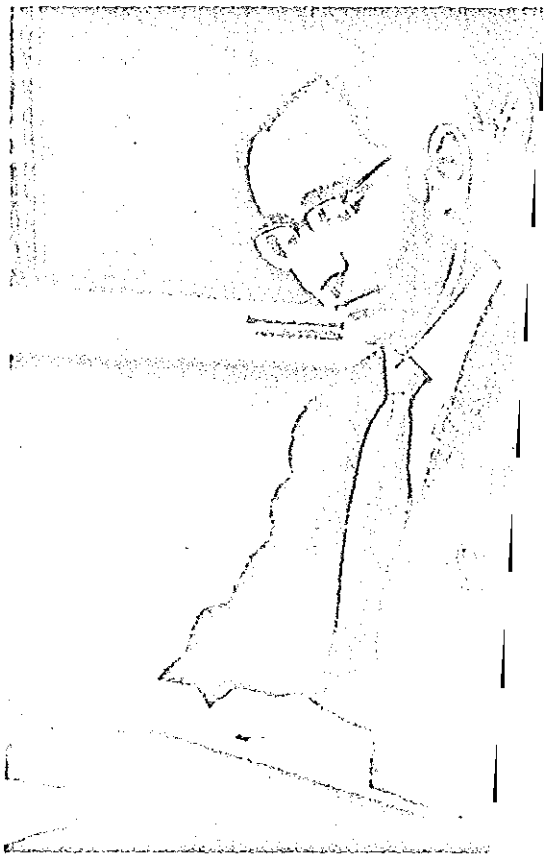
Philip W. Anderson

The discovery of what we have ever since called the "Josephson effect" took place while I was visiting Cambridge, England, for a sabbatical year in 1961–62. Recently, when I travelled to Kyoto to accept the London Prize on Brian Josephson's behalf, I put together some reminiscences of that period, which may be more interesting than a mere recapitulation of the bare scientific facts. Lest I appear too central to this account, I should make it clear that at least two other people could have told a similar story from their own points of view. They are Brian Pippard, Josephson's thesis advisor while the work was being done, and David Shoenberg, director at that time of the Mond Laboratory in which Josephson was a research student in experimental physics and where I served as nominal head of the solid-state theory group.

The discovery

Josephson had taken my course on solid-state and many-body theory. This was a disconcerting experience for a lecturer, I can assure you, because everything had to be right or he would come up and explain it to me after class. Probably because of the course and some of the things I said, he showed me his calculations within a day or two after first making them. At first he followed directly some recent calculations, by Morrel Cohen, Leo Falicov and James Phillips,[1] of the current by the "tunneling Hamiltonian" method, applied to two superconductors separated by an insulating layer. But by the time I saw the calculations he had already worked out the rather sophisticated formalism he later published in which he kept track of particle charges.

By this time I knew Josephson well enough that I would have accepted anything else he said on faith. However, he himself seemed dubious, so I spent an evening checking one of the terms that make up the current. We were all—Josephson, Pippard and myself, as well as various other people who also habitually sat at the Mond tea table and participated in the discussions of the next few weeks—very much puzzled by the meaning of the fact that the current depends on the phase. (This is the famous $J = J_1 \sin \varphi$ relation—see box on page 24.) I think it was residual uneasiness on this score that caused the two Brians (Pippard



Brian D. Josephson in 1969

Philip W. Anderson is at Bell Telephone Laboratories, Murray Hill, N.J.

The Josephson effect

Superconducting tunneling. Oxidise a strip of superconducting film (say lead) very lightly to a depth of 10–20 Å; then evaporate a cross strip of lead film. Simply plot the current–voltage characteristic of the resulting "tunnel junction." Electrons can tunnel quantum mechanically directly from an energy level in one piece of lead to an equal energy in the other. At temperatures less than T_c, the superconducting transition temperature, "normal" current does not flow very well until we apply enough voltage to overcome the energy gap 2Δ, because there are no single-particle levels in the gap.

Josephson effect. But Josephson showed that superconducting pairs of electrons can also tunnel, almost equally well. The supercurrent is $J = J_1 \sin\phi$, where J_1 is comparable with the "normal" current at $V = 2\Delta$. ϕ is the difference between the phases of the electron-pair "wave functions" in the two superconductors on opposite sides of the insulating barrier of oxide. Later it was realized that the phases are coupled by an energy $E = -E_1 \cos\phi$, that the minimum energy is achieved by equal phases and zero current. But an external current source can drive the phases unequal. If the phases are time independent, a dc supercurrent flows with no voltage drop. (See below.)

Superconductivity. The phenomenology of superconductivity can be summarized in terms of the phase ϕ of the pair wave functions. The superconducting electrons (usually most of them) move with a velocity v_s, which is given (except for a factor 2 in the vector-potential term) by the standard quantum-mechanical equation

$$v_s = \frac{\hbar}{m}\left(\nabla\phi - \frac{2e}{c}A\right)$$

If the phase is uniform, v_s is proportional to the vector potential A, which gives the standard London equation describing the Meissner effect of flux exclusion. The quantization of flux is just the requirement that ϕ be the phase of a single-valued function, so it may change only by $2n\pi$ or going around a ring. We see from this equation that either a magnetic field A or a current v_s can lead to a gradient of ϕ; either may be used in a Josephson interferometer.

The time dependence of the phase is given by the "Josephson equation"

$$\hbar\frac{d\phi}{dt} = 2eV$$

which is just the Einstein equation relating frequency to the energy of a pair. This equation, together with the previous one, shows that a voltage can not be maintained across a superconductor without accelerating the supercurrent. This equation, together with the Josephson-current equation, allows us to measure e/h from the relationship between voltage and the frequency of the ac Josephson current that flows when we maintain a voltage across the junction.

Josephson penetration depth. The Josephson current can itself cause a magnetic field that modulates ϕ. The resulting coupling leads to a modified kind of Meissner effect with a long characteristic length λ_J. Junctions larger than λ_J behave like bulk superconductors in some ways.

and Josephson) to decide to send the paper to *Physics Letters*,[2] which was just then starting publications, rather than to *Physical Review Letters*.

Earlier in my course I had made some remarks about broken symmetry, and later we discussed how broken symmetry made this peculiar behavior of the current possible. Josephson has always given me much more credit for these remarks than my understanding at the time deserved. Apart from these ideas about broken symmetry and some very minor points acknowledged explicitly in that *Physics Letters* paper, I want to emphasize that the whole achievement, from the conception to the explicit calculation to the publication, was completely Josephson's.

I hope it was no coincidence that these developments occurred in the thoughtful and stimulating atmosphere characteristic of the Cavendish. But the specific achievement was Josephson's own; this young man of 22 conceived this entire thing and carried it through to a successful conclusion.

Two things about that original paper have always struck me as remarkable. The first one is that, from the original idea of a dc supercurrent, he should immediately make the all-important leap not only to the ac supercurrent but also to the mathematics of how to synchronize it with an external ac signal. Furthermore, he explained how to observe the effect in exactly the way that Sidney Shapiro did nearly two years later,[3] and so predicted what is now the standard method for measuring e/h. The second remarkable thing was the initial response of our excellent patent lawyer at Bell Telephone Laboratories when John Rowell and I consulted him; in his opinion Josephson's paper was so complete that no one else was ever going to be very successful in patenting any substantial aspect of the Josephson effect.

Let me then complete the history. At this point I have to start looking at things from my own point of view, which was that I was extremely enthusiastic about what Josephson had done and eager to work on it. I believe I was probably the most enthusiastic evangelist for the effect that he had. Therefore this narrative will now begin to sound like a description of what I did with the Josephson effect. But of course at the same time other lines of development were being followed, lines I had nothing to do with.

After a few more discussions at the Cavendish, which did not carry us much further than what had been published in the first paper, I returned home to Bell Labs. There I mentioned to Rowell my conviction that Josephson was right. Rowell admitted to me that he had, from time to time, seen suggestive things in his tunnel junctions, and a few months later he called me in to look at his experimental results on a new batch of tunnel junctions. He thought he might actually be seeing the Josephson effect. In thinking at that time about the results of Rowell's measurements the penny finally dropped for me, and the only two remaining essential components of the phenomenon became clear.

The first of these is that $J = J_1 \sin\varphi$ implied a coupling energy between the two superconductors of $E = E_1 \cos\varphi$. The value of E_1 is a very vital point; it must be larger than the noise energy in the circuit, which is why our experiments with low-resistance junctions worked and previous attempts by Josephson and others with higher resistance junctions, and thus smaller J_1 and E_1, had not been successful. The second new idea was the Josephson penetration depth.

Looking back, I assume that many workers must have observed the coupling-energy effect previously, but the point is that this effect is not easy to distinguish from a tunnel junction shorted across the layer separating the two metals. We were able to see the effect because three conditions were satisfied: First, we knew what to look for; second, we understood what we saw. Both of these were the result of our contact with Josephson. The third condition was that we were confident of Rowell's skills in making good, clean, reliable, tunnel junctions.

After we understood these theoretical ideas we changed the title of our paper[4] from "Possible . . ." to "Probable Observation of the Josephson Effect."

Publication

The story of the publication of these various ideas is complicated. I soon learned that the two ideas I have just mentioned had both been discovered and written out by Josephson several months earlier. The history of how they were published is a kind of "Alphonse-Gaston" story.

The first time they were written down was in August 1962. Josephson submitted, in support of an application for a research fellowship at Trinity College, Cambridge, a "fellowship thesis" that contained the first really full account of the general nature and the physical meaning of the Josephson effect. This paper also contained the generalization to nontunneling situations, which I later called "weak superconductivity." One copy of this thesis remains, I believe, in the Trinity College library; one copy Josephson kept, and for some reason a photostat copy turned up in Chicago. Later on, after we did the experiments, I received a copy also. But the first three copies represented what Josephson felt to be adequate publication. Perhaps I am not being quite fair, because most of these ideas were included in Josephson's remarks at the LT8 conference in London, September 1962, in a rather famous debate with John Bardeen.

I had reproduced some of these results but, knowing after the fact about Josephson's thesis, I did not want to call too much attention to my own work of several months later. Therefore I published by own full version[5] in the notes of the Ravello Summer School of May 1963, which took almost two years to appear in print.

Neither of these two roughly equivalent papers is widely quoted, to say the least. That is a pity, because together they contain many things that had to be rediscovered later. I find in my file of letters from Josephson that we discussed the possibility of a joint paper, but this seems not to have come off.

Significance

This is enough reminiscence; what has turned out to be the significance of Josephson's discovery? I think two analogies may help to show how important it is. Imagine, for one, that we had developed a purely theoretical geophysical model of the earth that predicted the existence of a magnetic field, but imagine in addition that no one had yet invented the compass! I like that analogy because it shows up both aspects of the importance of a measuring instrument—and the Josephson effect is first and foremost a measuring instrument. The measuring instrument is important because, first, it verifies the theory, and second, it suggests a host of practical uses.

The second analogy for the significance of the Josephson effect is somewhat deeper. Suppose we understood the wave nature of light theoretically, and had even developed the laser, but no one had ever invented a way to get the light beam out without completely messing up the laser oscillations. That is, suppose no one had invented slits, half-silvered mirrors and all the other paraphernalia of interference experiments. We should then be in the same frustrating position that we were in with regard to superconductivity before Josephson. Then we had a theory and sources of coherent radiation, but no measuring instruments or gadgets to verify the theory and make use of the coherence.

In 1962 we had already postulated that superconductivity consisted of a coherence of the de Broglie waves representing pairs of electrons inside the superconductor. Prior to Josephson, the phase φ of these macroscopic waves was thought to be unmeasurable in principle, by the same kind of specious reasoning that has been used to prove such things as the nonexistence of ferroelectricity. In principle the phase varies with magnetic field and current in space according to the equation

velocity of superelectrons v_s

$$= \frac{\hbar}{m}\left(\nabla\varphi - \frac{2e}{c}\mathbf{A}\right)$$

and in time according to the Einstein–Josephson equation

$$\hbar\omega = \text{energy}$$

That is,

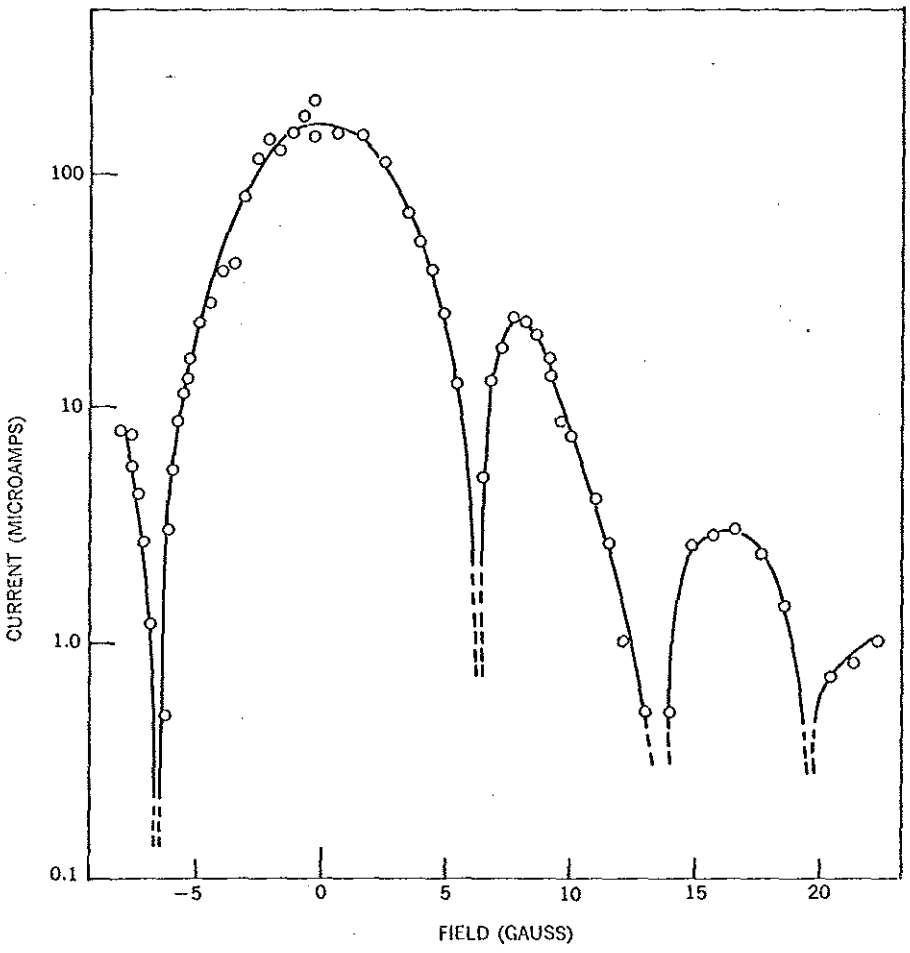
$$\hbar\frac{d\varphi}{dt} = 2\mu_e + 2eV$$

where μ_e is the chemical potential.

It was not yet entirely clear at the time that these two equations are equivalent to our theoretical understanding of superconductivity. The second equation predicts zero resistance in the absense of acceleration ($d\varphi/dt$), and the first leads to the Meissner effect and to flux quantization.

The four equations in the box on page 24 tell us all we ever need to know about the Josephson effect and most of what we need to know about superconductivity. The first two equations give us a way of comparing the phases of two weakly coupled bits of superconductor, if they are so weakly coupled that they do not seriously perturb one another.

The straightforward way to verify the Josephson effect is simply by observing the superconducting tunnel current between two bits of superconductor. It is important that the junction really be a tunnel junction. Josephson's theory predicts a certain magnitude for the tunnel current. We began[4] by making this observation, which is simple theoretically but not very elegant experi-



Single-slit interference pattern shows dependence of Josephson current in a lead–lead-oxide–lead junction on magnetic field at 1.3 K. From reference 6. Figure 1

mentally. A more convincing verification[6] is the observation of the single-slit interference pattern caused by a small, steady change in the magnetic field near the sample. The change in magnetic field gives a linear variation of phase across the sample in the same way that changing the angle of observation across a slit gives a linear change in phase across the slit.

Figure 1 shows an observation of this pattern, made by Rowell in 1963 with the best junctions he was at that time able to produce.[6] The extremely deep minima attest to the high uniformity of this junctions. The original observations of this effect, made by Rowell in December 1962, were comparatively crude. He simply observed the changes in the tunneling current on a recorder while he slid a bar magnet along the surface of a table towards the sample. Again from my file of correspondence with Josephson I note that he also had suggested this observation before he had any way of knowing that we were in the process of carrying it out.

In principle the next thing to do is the two-slit experiment. This was conceived and carried out[7] in many fascinating versions by the group of Robert Jaklevic, James Mercereau, Arnold Silver and John Lambe at the Ford Laboratories. Figure 2 shows their two-slit pattern, obtained as a function of v_s, which is proportional to the current through one arm of a loop containing a pair of junctions. In addition they observed the pattern in three other ways—as a function of the magnetic field alone, as a function of the vector potential in the absence of a field and as a function of the velocity of the superelectrons imparted through rotation.

The final coup verifying the Josephson effect was carried out[3] by Sidney Shapiro, of the A. D. Little Co, actually before the two-slit experiments were done. Shapiro applied an external ac signal to the junction and observed the points of synchronization with the internal ac supercurrent. Figure 3 shows the results of a similar experiment carried out later[8] with a thin-film bridge by Ali Dayem and myself; this geometry greatly enhances the coupling to external rf fields.

I do not think one should underestimate the impact of the Josephson discovery on our understanding of superfluidity and superconductivity. One example of great importance is in the study of dissipative phenomena. The idea and the experimental verification of flux and vortex quantization, and the Josephson frequency condition, already existed in the literature. It was the direct tangibility given them by the experimental access that made us put them together in the two phenomena, seemingly diverse, of high-field superconducting magnets and dissipation in superfluid helium.[9] In each case the key to what is going on in the dissipative effects is the Josephson frequency condition and the idea of phase slippage by vortex motion.

Applications

We have now reached the third stage of the application of Josephson's discovery, after a zeroth stage consisting of his predictions, a first stage of verification and a second stage of generalization and conceptualization. This third stage is practical application, which of course means scientific application in other fields as well.

At present by far the most famous application is the measurement of h/e, as suggested by Josephson himself very early and as carried out[10] by Donald Langenberg, Barry Taylor and William Parker of the University of Pennsylvania. The story of this measurement is too well known to bear repetition; I should say only that, from the first, all of us in the field have felt that it was only a matter of time before the most accurate standard of voltage measurement became the comparison with frequency via Josephson's equation. This role is analogous to that of light interferometry in length measurements.

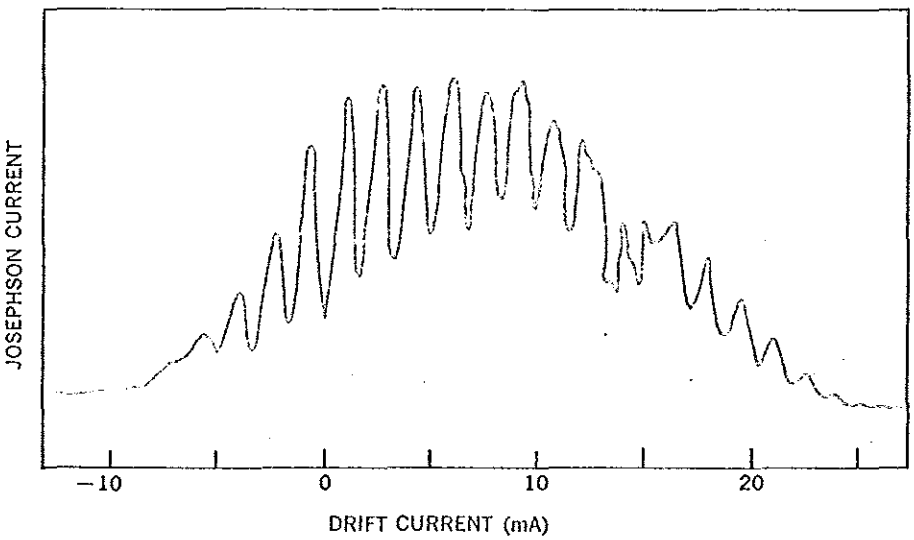
We can assume that h/m_e and h/m_{He}, two more constants that can also be measured by Josephson experiments, will also fall to sufficiently ingenious and careful experimentalists. The experiment for h/m_e, in particular, is very clean; one has only to count and to measure an area. Parker has already done this experiment to an accuracy of 10^{-4}. I see no reason why the major portion of our system of units should not be basically defined in the future by two kinds of interferometry—Josephson interferometry and light interferometry.

Perhaps even more important will be the role of the effect in ultrasensitive electromagnetic (and, with liquid helium, gravitational) measurements. The pioneering work of Jaklevic, Mercereau and their coworkers has been followed by the development, at the Mond Laboratory at Cambridge, of the SLUG picovoltmeter by John Clarke.[11] This instrument can measure voltages such as those that arise from the resistance of a 10-nanometer (100-angstrom) copper film normal to its area from one superconductor to another. I have no idea what further plans for ultrasensitive measurements are on the drawing board. But it is a bit frightening to realize that Clarke's voltmeter was a success because he deliberately reduced the sensitivity to make the instrument easier to handle!

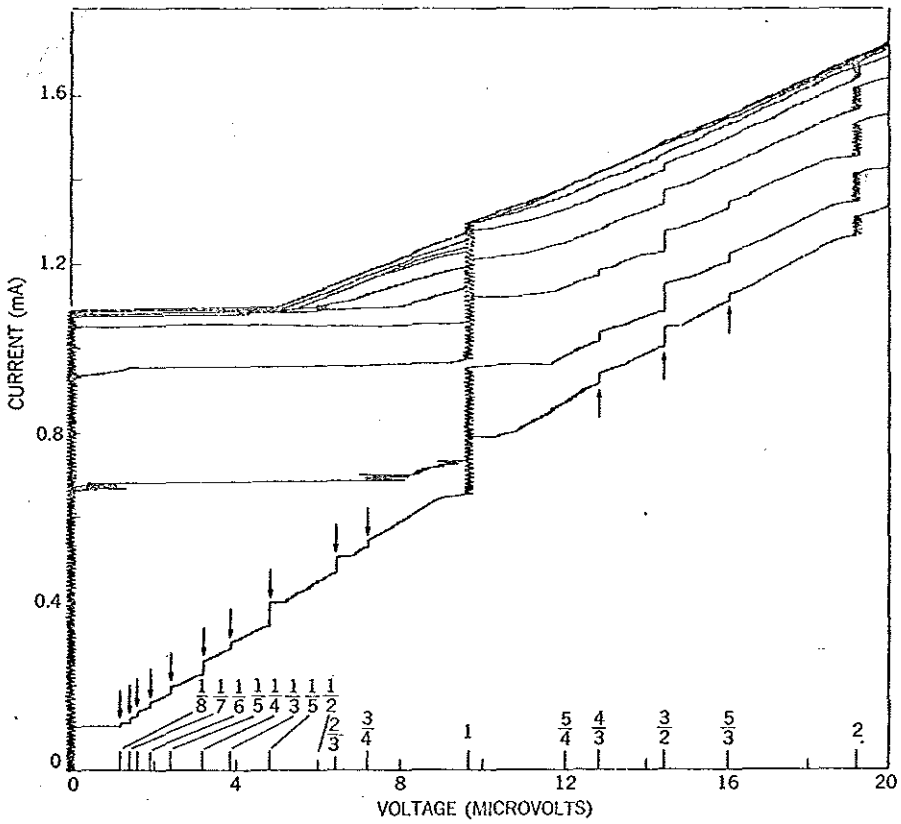
A third promising direction is in the detection and demodulation of high-frequency radiation, pioneered at Bell Labs by Dayem, Shapiro, Mike Grimes and Paul Richards.[12] So far the natural convenience of the ac Josephson effect as a direct frequency-to-voltage converter (the relation is directly linear) has not been fully exploited, but great promise exists.

The use of the Josephson effect as a high-frequency generator has been discussed widely, but I have a faint personal prejudice against such a use. It appears to me to be a crude application for this delicate effect. Of course I hope I shall be proved wrong and the ac generator will turn out to be highly valuable in the region between the microwave and the infrared frequencies.

Perhaps the wildest application yet is at the General Electric research labora-



Interference and diffraction effects in two thin tin films separated by an oxide layer, vacuum deposited on a quartz substrate in such a way as to give two junctions connected in parallel. This experimental trace of Josephson current versus drift current at 3.7 K shows the two-slit pattern. Maximum current is 1.5 mA. Zero offset arises from a static applied field. Figure 2



Effect of an external ac signal. These are current–voltage characteristics for a thin-film bridge of indium subjected to a 4.26-GHz external signal. The steps in the characteristic curves are induced at voltages $V = nh\nu/2me$. Fractions in color are values of n/m. The separate curves are for different power levels. From A. H. Dayem, J. J. Wiegand, Phys. Rev. *155*, 419 (1967). Figure 3

ries at Schenectady, where Ivar Giaever[13] has a light-sensitive Josephson junction that can be switched on with light.

An application I have been advocating for many years would involve using Josephson junctions for the control and movement of single quantized vortex lines as computer elements, rather like the recent "bubble" devices developed at Bell Labs. Rough calculations indicate great advantages in size and speed, but so far no real work has been done on such systems. The technology is challenging, to say the least.

With these last speculative comments I have tried to convey the impression that the Josephson effect is not a closed subject but still very much open-ended. The technology as well as the science of the future is likely to depend greatly on macroscopic quantum interference of matter waves, the discovery and exploitation of which we owe in large part to Brian Josephson.

* * *

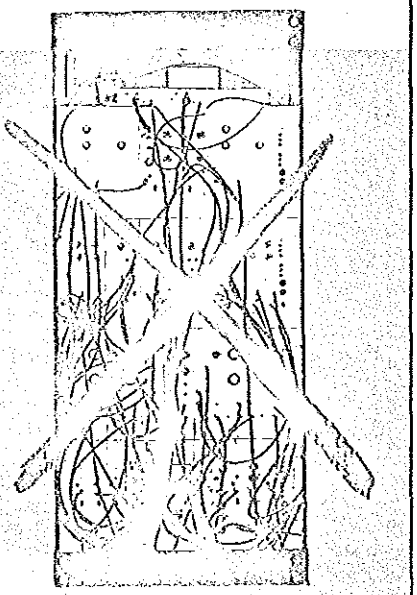
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